Certificate-based verification of the fundamental-group and framing steps in Wuebben's exotic $S^2 \times S^2$ construction

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Abstract

Wuebben (arXiv:2608.17267v1), building on Lidman–Piccirillo (arXiv:2505.14387v1), proves that a specified symplectic 4-manifold V with the homology of $S^2 \times D^2$ is simply connected and deduces three theorems: its symplectic double is an exotic $S^2 \times S^2$; its quotient is homeomorphic to Kawauchi's manifold B but distinguished from it by the smooth sliceness of the figure-eight knot; and the Lidman–Piccirillo regluing is an exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. We give a certificate-based verification of the theorem-critical fundamental-group and framing steps and a machine-replayed audit of the downstream argument. We rebuild the bundle as a simplicial complex from the paper's marked-fiber data, extract the surgery tori and based peripheral curves, and prove that the four convention-specialized filled presentations containing the paper's fixed filling are trivial. Each computation emits a derivation certificate accepted by independently implemented Python and Ruby checkers. Exact-rational Python programs verify the displayed calculus in the framing lemma identifying the combinatorial and Lagrangian longitudes and finite scripts check its stated combinatorial hypotheses. We then express the deductions to the three theorems as an explicit dependency chain: twenty-five external theorems are stated with their hypotheses; every hypothesis is discharged by a named certificate, computation, or earlier step; and two checkers replay every finite calculation. Thus the three theorems are verified relative to explicitly named external results, with no known project-specific gap. The theorems are Wuebben's; this note claims only the verification. All artifacts are public in a pinned release.

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1 Introduction

For the construction of [1], the theorem-critical step is the fundamental-group computation. The manifolds there rest on a specified Lidman–Piccirillo piece V, obtained by two Luttinger surgeries [5, 3] on Lagrangian tori in a symplectic non-product genus-2 surface bundle R over a once-punctured torus, following the framework of [2]. Write σ for the free orientation-reversing involution of $\partial V = S^3_0(Q)$, Q the square knot, $Z = V \cup_\sigma V$ for the symplectic double, $W = V/\sigma$ for the quotient, and $Z'' = V \cup_{f\circ\sigma} V$ for the Lidman–Piccirillo regluing. The paper's three theorems are: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to Kawauchi's manifold B, and (B, W) is a homeomorphic, non-diffeomorphic pair distinguished by the sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. All three rest on the simple connectivity of V. Wuebben fixes one geometric manifold: his $(m,n)=(0,0)$ member, using two geometric $+1$ Luttinger surgeries and the y_1-based Ax half-drift on T_α. The exponents ϵ_A, ϵ_B in his relation sheet are orientation and presentation conventions, not four geometric parametrizations; he allows either sign for each. We certify all four convention specializations, so in particular the relator sheet for the fixed V is included. The other three specializations are convention-robustness checks, not claims about three further manifolds. Wuebben's own 4,096-system grid varies, besides these two exponents, the three meridian signs of his transport corrections, the left/right placement of those corrections, the choice of conjugating arc for the Bs correction, and the arc, sign, and placement of the T_β push-off. In the rebuilt model none of these is a free choice: the corrections, basings, and push-off are computed as literal certified loops (Section 3), so only ϵ_A, ϵ_B remain as conventions, and the four certificates cover his grid at the fixed member.

This note reports an independent implementation of that step, a machine audit of the framing lemma that connects the group theory to the symplectic geometry, and an explicit proof chain from simple connectivity to the three theorems. Nothing in the implementation reuses the paper's coordinates, code, or intermediate data: the geometry was rebuilt from the paper's stated marked-fiber data alone, and the paper's displayed relations, sign tables, and Table 1 filling words were then *compared against* the rebuild rather than assumed. Every compared item agreed, and the independent raw-paper extractor found no paper-to-model dictionary discrepancy. The downstream deductions are Wuebben's, and this note reproves none of the classification, symplectic, or Floer-theoretic theorems they invoke; what it adds is that each deduction is written out, each theorem invoked is stated with its hypotheses, and each hypothesis is discharged by a named certificate or computation.

For the notation below, "$n=0$" means Wuebben's Ax half-drift label on T_α; the adjacent "$n=1$" control replaces Ax by Ar^{-1}. The letters M, N denote based meridians of T_α, T_β, while A, B are the two base generators.

The result separates into an unconditional computational theorem, an identification that connects it to the paper, and a chain relative to named inputs.

Theorem 1 (certified triviality)**.** *For $\epsilon = (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2$, let P_ϵ be the corresponding convention-specialized filled presentation of Section 4: the exported 4-generator, 95-relator presentation of the complement of the two surgery tori in the simplicial model, together with the two filling relators of the n=0 double surgery with relation-sheet exponents (ϵ_A, ϵ_B), imposed with the certified based meridians and framed longitudes. Each P_ϵ presents the trivial group. The proof consists of four derivation certificates, contained in release `v1.5.5` of the verification repository, which replay from a fresh checkout and are accepted by independently implemented Python and Ruby checkers.*

Lemma 2 (certificate soundness)**.** *A certificate is a finite sequence of records, each an equation between words in the generators and their inverses, of four kinds: an input relator of the hash-bound presentation; an inverse axiom $gg^{-1}=1$ of the free group; an overlap, the critical pair of two earlier records, in which each branch is reduced by a recorded trace of rewrites by earlier records; or a change, an earlier record whose two sides are each reduced by such a trace. An identity root for a generator g is an index into the sequence at which the record is $g \to 1$. The checker reads the generator count from the certificate, requires it to equal the count* *in the hash-bound input presentation, and then requires an identity root for each numbered generator and its inverse. A checker accepts only if every record is derived as stated from records preceding it. Every accepted record is then an equality holding in the group presented by the input relators, so acceptance implies that this group is trivial. Neither the confluence of the completed rewriting system nor the correctness of the program that produced it enters the argument.*

Proposition 3 (identification)**.** *For every coherent choice of the paper's orientation conventions, the corresponding specialization of Theorem 1 presents $\pi_1(V)$ for the single specified manifold. Thus the four labels express convention choices for one target, not four target manifolds: the simplicial model is identified with the paper's marked bundle by a constructed fiberwise homeomorphism (Section 2); its peripheral loops are based meridians and product-framed longitudes of the surgery tori (Section 3); and the product framing agrees with the Lagrangian framing of Luttinger surgery by the paper's Lemma 8.2, audited by the exact programs and geometric arguments described in Section 3.*

Theorem 4 (downstream implication, relative to named inputs)**.** *For the specified piece V, relative to the twenty-five external theorems listed in Section 6, the certificates of Theorem 1, the geometric identification of Proposition 3, and the dependency certificate of Section 6: (A) Z is homeomorphic but not diffeomorphic to $S^2 \times S^2$; (B) W is homeomorphic to B, and the pair (B, W) is homeomorphic but not diffeomorphic, distinguished by the smooth sliceness of the figure-eight knot; (C) Z'' is a closed simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. Every hypothesis of every external theorem is discharged by a named certificate, computation, or earlier step of the chain, and two independently implemented checkers replay the finite dependency bookkeeping and verify that each of (A), (B), (C) depends on Theorem 1.*

Theorem 1 is unconditional: it is a finite object that any reader can re-verify. Proposition 3 is where the residual risk of any machine verification lives, since a machine certifies properties of a model; it rests on the reading of the paper's conventions, on elementary inputs of PL topology, and on the named theorems of Section 7. Theorem 4 is relative in a different way: its external inputs are deep theorems that are cited, not reproved, and the verification consists in the exactness of the bookkeeping between them. The accurate summary of the status of the whole is: *the three theorems are verified relative to explicitly named external results, with no known project-specific gap* — a statement about verification depth, offered to facilitate expert review, not to replace it.

Two disclosures. The theorems are Wuebben's; this note asserts nothing about priority or authorship of the mathematics and claims only the verification. The verification was carried out by Anthropic Claude models (Fable 5 and Opus 5) and OpenAI Codex (GPT-5 family) under the author's direction, with commit-level provenance recorded in the repository's ledger and corrections from cross-model review preserved. Mathematical responsibility rests with the human author. Model provenance does not itself establish correctness: readers should inspect the public checkers and replay their certificates.

2 The independent model and the interpretation dictionary

The genus-2 fiber is realized as a regular neighborhood of the paper's five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0

realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

The identification with the paper's marked bundle is by literal based monodromy action: the based mapping classes of both monodromies are certified as equalities of vertex paths, transported by proof-producing Tietze chains that replay step by step, and completed by an explicit marked mapping-cylinder clutching over the base graph. The identification is thus a constructed fiberwise homeomorphism into the paper's bundle rather than an appeal to the classification of surface bundles. Everything the surgery argument needs is transported through that map into the paper's already-smooth manifold, where the smooth and symplectic arguments run; no smoothing of the triangulation is ever chosen.

Because the machine certifies properties of a model, the model itself is the residual risk. Five structurally different implementations reconstruct it independently, arranged so that each class of translation error — subdivision artifacts, monodromy transcription, ordering, sign, and orientation conventions — would be caught by at least one of them. What redundancy cannot exclude, a shared misreading of the paper's conventions, is addressed by a bound interpretation dictionary: every convention read from the paper carries the certified computation that would fail under a rival reading. Wuebben's committed scripts enter at exactly one point, parsed and never executed: after the dictionary was frozen, selected entries were compared diagnostically against them, and no verified conclusion consumes script-derived data. All selected entries agreed. The remaining transcription-level entries are audited by a quarantined extractor against the hash-pinned immutable Lidman–Piccirillo v1 arXiv source [2], in both its prose and its vector-figure layers; the one interpretation remaining at that boundary is the ordinary convention that dashed arcs in a handle picture depict hidden projection rather than extra crossings. That extractor distinguished the nearest rival twist-order and twist-sign readings and found no discrepancy with the independently built model. The full inventory of implementations and the dictionary are in the repository.

3 Peripheral data and the framing identification

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction package (the corrected relations of its §8.3 and the push-off basing correction of its §8.5) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper's corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The closed section $\widehat{\Gamma}$ of the doubled bundle — the union of the two copies of the section through a fixed point of φ_0 — is extracted as a closed orientable genus-2 surface, and its self-intersection $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ is certified with an explicit normal push-off; this number is used directly in Section 6.

Luttinger surgery is performed with respect to the Lagrangian framing; the paper's Lemma 8.2 identifies it with the fibered framing carried by the combinatorial longitudes above. This lemma carries an exact logical load: if the identification failed by j meridians, the certificates of Theorem 1 would remain correct for the product-framed presentations they literally describe, but would no longer identify those presentations with the paper's surgeries. The lemma's displayed Weinstein-chart, seam, and constant-momentum algebra are checked by exact-rational Python programs, and its inline analytic normalizations are rebuilt constructively rather than cited: the relative Moser flow is produced as the explicit monotone inverse of a cumulative coordinate, and the equivariant step defines the Moser field directly on the double cover with certified projection and deck invariance, so neither an ODE existence theorem nor a covering-lift theorem is invoked. Chart independence is proved between the paper's displayed charts by two routes, with the one symplectic move that could introduce meridian components — the closed-1-form shift — excluded by the zero-section condition; [3] (§2.1 and Proposition 2.2) stands as corroboration. These programs verify the displayed identities and the finite hypotheses of the local constructions; they are not a foundational formalization of differential topology.

Independently of the lemma, a finite scan tests what a framing error would do. It refills the complement with longitudes shifted by (j_a, j_b) meridians on (T_α, T_β) for the eight axis shifts $(j_a, 0)$, $(0, j_b)$ with $0 < |j| \leq 2$ and the four diagonal shifts $(\pm 1, \pm 1)$ — a cross-and-diagonal sample, not the full grid — across all four sign pairs and both half-drift families, giving 96 counterfactual presentations, plus four zero-shift controls that re-derive the paper's own n=0 fillings through the same pipeline. The completed scan reports triviality for all 100, but at two evidentiary levels: 22 cases have independently replayed derivation certificates (the 18 former holdouts plus the four zero-shift controls), whereas the other 78 have only the retained GAP session verdicts described below. No nontriviality witness was found at any stage. The scan pipeline, which simplifies each presentation before attempting completion, decided 82 cases at the level of a GAP session — 40 by Tietze collapse of the simplified presentation and 42 by its confluent completion — and those runs exported no scan-specific certificates; the four zero-shift controls among them are the paper's own fillings, whose separate certificates are those of Theorem 1. The 18 it could not decide, every one involving a shift on T_α, carry derivation certificates of the kind in Section 5: 17 from confluent completion of the unsimplified 97-relator presentation, whose redundant relators are what make completion tractable, and the last, the n=0 $(+,+)$ filling shifted by $(+1,+1)$, from a certified reduction through the 96 relators it shares with its n=1 companion, which complete confluently to an infinite cyclic group in which the omitted relator reduces to a generator. Within the scanned range, then, no framing error would have produced a nontrivial group. The scan is robustness evidence only: the paper's own fillings are the certified controls, and it bears on the proof only if Lemma 8.2 first fails. The table, methods, and search history are in the repository.

4 Certified triviality of the four convention-specialized groups

Filling the triangulation-derived complement presentation (4 generators, 95 relators) with the certified coherent peripheral pairs yields, for all four n=0 sign pairs, complete confluent rewriting systems in which every generator reduces to the identity. Each completion is exported as a derivation certificate, and Theorem 1 is the statement that the four certificates verify.

system	signs	derivation records
n=0	$(-,-)$	1123
n=0	$(-,+)$	1075
n=0	$(+,-)$	804
n=0	$(+,+)$	1504

Derivation-DAG certificates for the four convention-specialized filled groups; each verifies triviality from a fresh clone of the repository.

Four adjacent n=1 half-drift control presentations, obtained by replacing Ax with the adjacent minimal representative $Ar^{-1} = Ax(rx)^{-1}$ — Wuebben's n=1 member — also pass the identical certification. They verify his one-step re-indexing and carry no logical load for the fixed n=0 target; their certificates sit beside these in the same manifest.

5 Certificate verification

Knuth–Bendix completion (KBMAG 1.5.9 [22], run under GAP 4.11.1 [23]) is used only as an *untrusted certificate generator*. A patched build of KBMAG records the ancestry of every equation it derives; the patch is distributed as `verification/luttinger/kbmag-proof/kbfns-proof.patch`. An untrusted compiler prunes that history to the dependency cone of the final identity rules and emits a certificate of the form in Lemma 2, binding the input presentation by SHA-256. The certified set is enforced, not assumed: both checkers reject a batch that verifies any case twice, and in full-inventory mode — the documented invocation — require the batch to be exactly the input's eight fillings, one file named by each case slug, over a source with four generators, 95 complement relators, and two filling relators per case; the manifest generator refuses to hash a certificate directory that is not exactly one file per filling. A successful replay therefore verifies exactly the four n=0 sign pairs and the four n=1 controls. A small independent checker then re-verifies every input-relator axiom, inverse axiom, critical overlap, rewrite trace, tidying change, and final generator-to-identity rule; it neither imports nor runs KBMAG. A second checker, written independently in Ruby, re-verifies all retained derivation records against the same hash-bound certificates and accepts, importing neither the first checker nor any project group code. All certificates, the exported rewriting systems, and a hash manifest replay from a clean checkout, and the stored rewriting systems reload and re-verify under GAP/KBMAG independently of the certificate path. The design principle throughout is that certificates are checked, not trusted: search and completion tools are untrusted generators, and every computational claim rests on a replayable artifact validated by small independent code.

The machinery does not only produce trivial groups, and the checkers do not only accept. Abelianization is only a sanity check here: the 95-relator complement presentation has $H_1 = \mathbb{Z}^2$, and each pair of filling relators kills it, so every filled presentation has trivial abelianization; that the filled groups are trivial rather than merely perfect is carried entirely by the certificates. The same pipeline, run as a calibration on the standard T^4 Luttinger example, returns the complement group $\mathbb{Z}^2 \times F_2$ and the textbook surgery quotient $\mathbb{Z}^4$; on a Baldridge–Kirk example it separates the two sign conventions by the trace fingerprints 1 and 3, never their negatives; the 96-relator common core of Section 3 completes to an infinite cyclic group; and a mapping-torus control during the fiber construction reported $H_1 = \mathbb{Z}^2$ against $\mathbb{Z}^3$ for the identity, which is what exposed the one construction error found and fixed in the project. On the checker side, four deliberate corruptions of an accepted certificate — an identity root pointed at the wrong record, an altered input-relator equation, an altered rewrite trace, and a wrong generator count — are each rejected, as is a batch that supplies the same certificate twice.

Both filled-group checkers were produced by OpenAI Codex (GPT-5 family), as separate Python and Ruby implementations. They use different languages and share no implementation code, but they do share the published certificate specification and input format; "independent" here means implementation-independent, not cross-model authorship or a claim that model-generated software has statistically uncorrelated errors. A Claude Fable 5 session independently reconstructed and replayed the 96-relator common-core computation behind the last framing-scan case of Section 3; that was a cross-model replay of one case, not authorship of either checker. The code, not that provenance, is the object offered for review. No claim is made that either checker has yet received an independent human line-by-line audit.

6 From simple connectivity to the three theorems

The deductions from $\pi_1(V) = 1$ to Theorems A, B, C occupy Part I of [1], whose Theorem 1.2 reduces them to the simple connectivity of V. This section carries those deductions through as an explicit dependency chain. The chain is itself a machine artifact: a script records every item as one of four kinds — an *external theorem* stated with its hypotheses and source; a *certificate* of this repository, bound by SHA-256; a *computed* fact, executed by the script; or a *step*, a deduction from earlier items — and freezes the result as a certificate. A second script, written independently in Ruby, re-derives every computed fact from scratch, checks that every citation resolves and that the dependency graph is acyclic, verifies that each of the three conclusions depends on Theorem 1, and checks every evidence file against its recorded digest. The chain has 25 external theorems, 3 certificates, 15 computed facts, and 16 steps; its prose form is in the repository, and what follows is the mathematics, with the inputs first. Both scripts in this downstream pair were produced by Claude Fable 5; independence here likewise means separate implementations, not cross-model authorship.

External inputs

Elementary algebraic topology enters as: the Seifert–van Kampen theorem for a union along a connected bicollared subspace; the exact sequence $1 \to \pi_1(Z) \to \pi_1(W) \to G \to 1$ of a connected regular covering with deck group G; Poincaré–Lefschetz duality and universal coefficients, with the Euler characteristic as an alternating sum of Betti numbers over any field, multiplicative for bundles with compact fiber and additive along a common boundary; the index formula $\det(\mathrm{Gram}\, S) = [L : S]^2 \det(\mathrm{Gram}\, L)$ for a finite-index sublattice, with the unimodularity of the intersection form on H_2/Tors; Wu's formula $\langle w_2, x\rangle = x \cdot x$ and its consequence $\langle w_2(X), [F]\rangle = \chi(F) + F \cdot F \pmod 2$ for an embedded oriented surface; Novikov additivity of the signature; asphericity of bundles with aspherical fiber and base; the adjunction formula $K \cdot S + S \cdot S = 2g - 2$ for a symplectic surface; and the isotopy of any two orientation-preserving smooth embeddings of B^4 in a connected 4-manifold [21], which together with the amphichirality of the figure-eight knot makes smooth sliceness in a closed 4-manifold a diffeomorphism invariant.

The named theorems are the following.

- **Freedman** [6]: closed simply connected topological 4-manifolds are classified up to homeomorphism by the intersection form and the Kirby–Siebenmann invariant; every unimodular form is realized, uniquely if even (with KS $\equiv \sigma/8$), and by exactly two manifolds if odd, one per value of KS.

- **Hambleton–Kreck** [7, 8, 9], in the form of [10, Theorem 5.1]: "Closed, oriented topological 4-manifolds with finite cyclic fundamental groups are classified up to homeomorphism by $\pi_1(M)$, q_M, the w_2-type, and KS(M)," where q_M is the form on H_2/Tors and the w_2-type is (II) for spin M.

- **Thurston** [13] and the paper's Thurston-type form ([1], proofs of Proposition 1.3 and Lemma 8.2; [2], proof of Theorem 8): a closed surface bundle over a closed surface with homologically essential fiber is symplectic, and for $R \cup_\sigma R$ the form can be chosen positive on the fiber F and on the closed section $\widehat{\Gamma}$, with the surgery tori Lagrangian.

- **Luttinger surgery** [5, 3]: surgery on a Lagrangian torus yields a symplectic manifold, agreeing with the original outside the surgered neighborhood and with unchanged Euler characteristic.

- **Symplectic Kodaira dimension** [12], as summarized in [11, p. 1]: κ is defined from a minimal model and "is independent of the choice of symplectic form ω"; a minimal symplectic 4-manifold has $\kappa = -\infty$ iff it is rational or ruled, and rational and ruled 4-manifolds have $\pi_2 \neq 0$. **Ho–Li** [11, Theorem 1.1]: "The Luttinger surgery preserves the symplectic Kodaira dimension."

- **Symplectic Thom** [14]: an embedded symplectic surface in a closed symplectic 4-manifold is genus-minimizing in its homology class; and the construction ([1], proof of Lemma 4.3, after [17]) of a connected symplectic unbranched k-fold cover of a square-zero symplectic surface inside its tubular neighborhood, representing k times its class.

- **Klug** [18, Theorem 2]: "Let X^4 be a smooth compact connected oriented 4-manifold with ∂X an integer homology sphere. Let F^2 be an orientable characteristic surface with connected boundary that is properly embedded in X. Then $\mathrm{Arf}(F)+\mathrm{Arf}(\partial F) = (\sigma(X)-[F]^2)/8+\mu(\partial X) \pmod 2$." For spin X, characteristic means vanishing in $H_2(X, \partial X; \mathbb{Z}/2)$. **Levine** [19]: $\mathrm{Arf}(K) = 0$ iff $\Delta_K(-1) \equiv \pm 1 \pmod 8$.

- **Slice disks and the 0-trace** ([2], proof of Theorem 1): a smooth slice disk with Seifert-framed normal bundle embeds the simply connected 0-trace, in which a capped Seifert surface is a closed square-zero surface generating H_2; the framing condition is automatic when the relative H_2 is torsion; the figure-eight knot has genus 1.

- **Lidman–Piccirillo's constructions** [2]: σ is a free orientation-reversing bundle involution (Lemma 4); the sections from the fixed points of φ_0 survive into V, their boundary circles are preserved by σ, and $R \cup_\sigma R$ is a closed genus-2 bundle over a closed genus-2 surface on which the surgeries act by Luttinger surgery on four disjoint tori missing F and $\widehat{\Gamma}$ (Lemma 6, proof of Theorem 8); $W = V/\sigma$ is a closed smooth oriented 4-manifold, double covered by Z, and is spin by the hyperbolic-pair argument of Lemma 7, whose hypotheses are the homology of V; the regluing diffeomorphism

f of $S^3_0(Q)$ exists, and with $\mathcal{A} \subset S^3_0(Q)$ the framed annulus joining $\partial\Gamma$ to its image under $f \circ \sigma$, the closed surface $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma$ has odd square and meets F once (proof of Theorem 2); and the Floer-theoretic input, Lemmas 9 and 10 with the vanishing of all mixed invariants of $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ (and of its sum with any homology 4-sphere) along both square-zero lines, via splittings over $S^2 \times S^1$ (proof of Theorem 2, using [15, 16]).

- **Kawauchi's manifold** [20, 2]: B is a closed smooth oriented spin 4-manifold with $\pi_1(B) = H_1(B) = \mathbb{Z}/2$ and $b_2(B) = 0$, in which the figure-eight knot is smoothly slice.

The chain

Throughout, F is a fiber of R disjoint from the surgery tori, Γ a section through a fixed point of φ_0, $\widehat{\Gamma} = \Gamma \cup_\sigma \Gamma \subset Z$ and $\Gamma'' = \Gamma \cup_{\mathcal{A}} \Gamma \subset Z''$ its closures, $\mathcal{A}$ the framed annulus above. The finite calculations cited are those of the chain's certificate.

Proposition 5. *$H_1(V) = 0$, $H_3(V) = 0$, $H_2(V) = \mathbb{Z}\langle F\rangle$ with $F \cdot F = 0$, and V is spin. Moreover $\pi_1(Z) = \pi_1(Z'') = 1$.*

Proof. Wuebben's Lemma 4.2 and the two filling relations give $H_1(V) = 0$ independently of simple connectivity; the certificate also implies it. The boundary $S^3_0(Q)$ is connected, so $H^1(V, \partial V) = 0$ and hence $H_3(V) = 0$, and $H_1(V, \partial V) = 0$ so $H_2(V)$ is torsion-free. $\chi(V) = \chi(R) = \chi(F)\chi(\text{base}) = (-2)(-1) = 2$, unchanged by the surgeries, and $b_0 = 1$, $b_1 = b_3 = b_4 = 0$ give $b_2(V) = 1$. A section is a properly embedded surface meeting F once, so $[F]$ is primitive and generates. $F \cdot F = 0$ since a fiber has trivial normal bundle. $H_2(V; \mathbb{Z}/2) = \mathbb{Z}/2\langle F\rangle$, w_2 is detected on it, and $\langle w_2, F\rangle = \chi(F) + F \cdot F = -2 + 0 \equiv 0$. Finally Z and Z'' are unions of two copies of V along the connected bicollared boundary, so by van Kampen each fundamental group is a quotient of $\pi_1(V) * \pi_1(V) = 1$. This is the one downstream step where simple connectivity of V, rather than only of Z, is needed: the regluing twists the amalgam. □

Proposition 6. *$H_2(Z) = \mathbb{Z}\langle F, \widehat{\Gamma}\rangle$ with intersection form the hyperbolic form H; Z is closed, simply connected, spin, with $b_2 = 2$ and signature 0; and Z is homeomorphic to $S^2 \times S^2$.*

Proof. $\chi(Z) = 2\chi(V) - \chi(S^3_0(Q)) = 4$ and $\pi_1(Z) = 1$ give $b_2(Z) = 2$ with $H_2(Z)$ torsion-free. $F \cdot F = 0$, $F \cdot \widehat{\Gamma} = 1$ (a fiber meets a section once), and $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ by the certified self-intersection computation of Section 3. The Gram determinant is -1 and the form is unimodular, so the index formula makes $(F, \widehat{\Gamma})$ a basis and the form is H. H is even, so $w_2(Z) = 0$ by Wu's formula. For even forms $\mathrm{KS} \equiv \sigma/8 = 0$ is determined, so Freedman's classification gives $Z \cong S^2 \times S^2$. Note that spinness of Z is obtained here from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$; the paper's route through the spin quotient W is not needed for this step. □

Proposition 7. *Z is a closed symplectic 4-manifold in which F and $\widehat{\Gamma}$ are symplectic surfaces of genus 2 and square 0, and Z is not diffeomorphic to $S^2 \times S^2$.*

Proof. $R \cup_\sigma R$ is a closed surface bundle whose fiber is homologically essential ($F \cdot \widehat{\Gamma} = 1$), so it carries the Thurston-type form, positive on F and $\widehat{\Gamma}$. The surgery tori are Lagrangian for it and the surgeries are the certified Luttinger surgeries (Proposition 3), and Luttinger surgery preserves the form away from the surgered neighborhoods, which miss F and $\widehat{\Gamma}$. For the non-diffeomorphism: $R \cup_\sigma R$ is aspherical, hence minimal (an exceptional sphere would be null-homotopic, hence of square 0, not -1) and neither rational nor ruled (those have

$\pi_2 \neq 0$), so $\kappa(R \cup_\sigma R) \neq -\infty$; Luttinger surgery preserves κ, so $\kappa(Z) \neq -\infty$. If $\psi\colon Z \to S^2 \times S^2$ were a diffeomorphism, composing with a reflection of one factor if necessary makes it orientation-preserving; ψ^* of a product form is then a symplectic form on Z inducing its orientation with $\kappa = \kappa(S^2 \times S^2) = -\infty$, contradicting the independence of κ from the form. $\square$

Propositions 6 and 7 are Theorem A.

Proposition 8. *W is a closed oriented smooth spin 4-manifold with $\pi_1(W) = \mathbb{Z}/2$, $b_2(W) = 0$, signature 0, and $\mathrm{KS}(W) = 0$; and W is homeomorphic to B.*

Proof. $Z \to W$ is a connected 2-fold covering, so $|\pi_1(W)| = 2$. $\chi(W) = \chi(Z)/2 = 2$ and $b_1(W) = 0$ give $b_2(W) = 0$, hence signature 0. W is spin by Lidman–Piccirillo's Lemma 7, whose hypotheses Proposition 5 supplies, and $\mathrm{KS}(W) = 0$ since W is smooth. W and B are then closed oriented topological 4-manifolds with the same Hambleton–Kreck invariants: $\pi_1 = \mathbb{Z}/2$, the zero form on $H_2/\mathrm{Tors} = 0$, w_2-type (II), $\mathrm{KS} = 0$. $\square$

Proposition 9. *No nonzero square-zero class in $H_2(Z;\mathbb{Z})$ is represented by a smoothly embedded torus.*

Proof. For $k \geq 1$ the connected symplectic k-fold covers of F and of $\widehat{\Gamma}$ represent kF and $k\widehat{\Gamma}$ and have genus $k+1$, since $\chi = -2k$; by the symplectic Thom theorem these genera are minimal, and reversing orientation covers $k < 0$. Since $(aF + b\widehat{\Gamma})^2 = 2ab$, every nonzero square-zero class is kF or $k\widehat{\Gamma}$ with $k \neq 0$, of minimal genus $|k| + 1 \geq 2$. $\square$

Proposition 10. *The figure-eight knot is not smoothly slice in W.*

Proof. Let D be a smooth slice disk in $W^\circ = W \setminus B^4$. The group $H_2(W^\circ, \partial; \mathbb{Z}) \cong H^2(W) \cong \mathbb{Z}/2$ is torsion, so the relative self-intersection of D is 0 and its framing is the Seifert framing. If $[D, \partial D] = 0$ in $H_2(W^\circ, \partial; \mathbb{Z}/2)$, then D is characteristic in the spin manifold W°, and Klug's theorem with $\mathrm{Arf}(D) = 0$, $\sigma(W^\circ) = 0$, $[D]^2 = 0$, $\mu(S^3) = 0$ forces $\mathrm{Arf}(4_1) = 0$; but $\Delta_{4_1}(t) = -t^2 + 3t - 1$ gives $\Delta_{4_1}(-1) = -5 \equiv 3 \pmod 8$, so $\mathrm{Arf}(4_1) = 1$. Otherwise the 0-trace $X_0(4_1)$ embeds in W and the torus T obtained by capping a Seifert surface with D has square 0 and $[T] \neq 0$ in $H_2(W;\mathbb{Z}/2)$. Since $X_0(4_1)$ is simply connected it lifts to Z, and T lifts to a torus $\tilde{T}$ with $p_*[\tilde{T}] = [T] \neq 0$ in $\mathbb{Z}/2$ coefficients; so $[\tilde{T}] \neq 0$ in $H_2(Z;\mathbb{Z}/2) = H_2(Z;\mathbb{Z}) \otimes \mathbb{Z}/2$ (as $H_1(Z) = 0$), and a class with nonzero mod-2 reduction is nonzero in the torsion-free $H_2(Z;\mathbb{Z})$; and $\tilde{T} \cdot \tilde{T} = T \cdot T = 0$ because p is a local diffeomorphism. This contradicts Proposition 9. $\square$

Proposition 8, Proposition 10, the sliceness of the figure-eight knot in B, and the diffeomorphism invariance of sliceness give Theorem B.

Proposition 11. *Z'' is closed, simply connected, smooth, with $b_2 = 2$, signature 0, and odd intersection form $\langle 1 \rangle \oplus \langle -1 \rangle$; hence Z'' is homeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

Proof. $\pi_1(Z'') = 1$ and $\chi(Z'') = 4$ give $b_2 = 2$ with H_2 free. Both Z and Z'' decompose into the same two oriented copies of V along their common boundary, so Novikov additivity gives $\sigma(Z'') = \sigma(Z) = 0$, independently of the regluing map. $F \cdot F = 0$, $F \cdot \Gamma'' = 1$, and $\Gamma'' \cdot \Gamma'' = 2\ell + 1$ is odd; the Gram determinant is -1, so (F, Γ'') is a basis, and $E = \Gamma'' - \ell F$, $D = F - E$ satisfy $E \cdot E = 1$, $D \cdot D = -1$, $E \cdot D = 0$ for every $\ell \in \mathbb{Z}$: the form is $\langle 1 \rangle \oplus \langle -1 \rangle$. Of Freedman's two manifolds with this form the one with $\mathrm{KS} = 0$ — Z'' is smooth — is $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. $\square$

Proposition 12. *Z'' is not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$.*

Proof. Z is symplectic with $|K_Z \cdot F| = 2$ by adjunction (F symplectic of genus 2 and square 0) and $H^2(V) = \mathbb{Z}$, so Lidman–Piccirillo's Lemma 9 gives $\Psi_{V,t} \neq 0$ for both spin^c structures with $|\langle c_1(t), F\rangle| = 2$. Z'' is $V \cup V$ along $S^3_0(Q)$ with t restricting to the non-torsion $s_\pm$, $b_2^+(Z'') = 1$, and $\mathrm{span}\{F\}$ a square-zero line, so Lemma 10 gives a nonvanishing mixed invariant $\Phi_{Z'',\mathrm{span}\{F\},t}$. A diffeomorphism to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ would carry $\mathrm{span}\{F\}$ to one of its two square-zero lines — in $\langle 1\rangle \oplus \langle -1\rangle$, $a^2 - b^2 = 0$ forces $a = \pm b$ — along each of which the manifold splits over $S^2 \times S^1$ and every mixed invariant vanishes. $\square$

Propositions 5, 11 and 12 are Theorem C, and Theorem 4 is proved.

Three points where the chain's route differs from the paper's are worth recording. $H_1(V) = 0$ is taken from the certificate rather than from the paper's Lemma 4.2 and the surgery relations, which are needed only for the "forcing" direction of its Proposition 1.3. The intersection form of Z is identified from the certified $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$, so that Z is spin because its form is even; W spin still enters, for the Hambleton–Kreck invariants and for Klug's theorem. And the sliceness dichotomy is taken over $\mathbb{Z}/2$ throughout, which is exactly what makes the null-homologous case Klug's characteristic case and the lifted torus nonzero integrally.

7 Trust boundary and data availability

Inputs to Theorem 1 and Proposition 3

Theorem 1 has no mathematical input beyond Lemma 2; its trust rests on the two small checkers. Proposition 3 rests on elementary inputs of PL topology retained as standard without a specific citation — the ribbon-thickening and bistellar-trace interpretations that read the paper's figures into complexes, the classification of surfaces, the collapsible-3-ball and cyclic-knot-unknot criteria behind the local-flatness certificates, and simplicial fundamental-group presentation with Tietze theory — together with the Kerékjártó periodic-disk involution theorem [4] behind the disk extension of the paper's Lemma 7.1, and [3] as corroboration of the framing identification. The ribbon-interpretation input is narrower than the phrase suggests: the certified classification of equivariant ribbon rotations pins the implication from the paper's stated marked five-chain data to the equivariant normal form, and the stated data themselves are audited in the original Lidman–Piccirillo source, so what this boundary retains is the thickening of that data into complexes together with the dashed-arc convention of Section 2. This note reproves none of these.

Inputs to Theorem 4

The twenty-five external theorems of Section 6, each stated there with its hypotheses. The deep ones are Freedman's classification, the Hambleton–Kreck classification, the invariance theorems for symplectic Kodaira dimension (Li, Ho–Li), the symplectic Thom theorem, Klug's relative Rochlin theorem, and the Floer-theoretic lemmas of Lidman–Piccirillo with the Ozsváth–Szabó nonvanishing theorem behind them; the rest are elementary or are constructions of [2] that this note takes as stated. None is reproved. The repository's dependency ledger records the whole verification as an acyclic graph of 33 named external theorems, 32 machine certificates, 11 geometric arguments, and 18 derived claims ending in the three theorems, with every evidence file bound by hash. These are not competing counts: the 25/3/15/16 figures in Section 6 describe only the compact downstream chain, while 33/32/11/18 describe the larger project-wide ledger that also contains the model, peripheral, framing, and PL verification.

Data availability

This verification targets Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`

The target-version ledger in `TARGET.md` also records the Lidman–Piccirillo v1 digest. All permanent certificates, frozen inputs, replay scripts, hash manifests, run transcripts, and provenance records are contained, with clean-checkout replay instructions, in the tagged release `v1.5.5` of the repository: `https://github.com/VentiMath/exotic-s2xs2-verification/releases/tag/v1.5.5`. The `verification/` tree of that release is identical to that of the repository commit

`d7152d269c54bb60d85fa1a11da09530b75c3119`

its filled-group proof manifest has SHA-256

`0c9993613e8b2099e7da0b15369ec0b7c9c575bf28ce751a4e72dc60e66a6fb2`

and its downstream-chain certificate has SHA-256

`ef2c5cd7c4824771236dab4df56ae57017c108d9e689432325600319388d5cb0`

The repository itself is public at `https://github.com/VentiMath/exotic-s2xs2-verification`; the versioned release link above, rather than search-engine indexing, is the artifact to check. The raw completion histories from which certificates are compiled are reproducible build products and are omitted. The paper's own code is at `https://github.com/bwuebben/exotic-s2xs2`; nothing in the verification derives from it except where explicitly cited.

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